

BROKILD · WINDOWS VST3 · HI-HAT AND CYMBAL SYNTHESISER

HATS OFF

A plate of bronze, two of them on a pedal, and the circuits that copied them. Every hat and cymbal, from those. No samples.

MANUAL · BUILD 260926.4

0.00 s0.50 s1.00 s1.50 s2.00 s2.50 s3.00 s3.50 s

bell: pingbow: rideedge: crash

CLOSEDPEDALOPEN

METAL

PITCHSIZEBRONZEDENSITYBLOOMTRASHDECAYNOISE

0.0 st18.0 in100 %95 %70 %0 %4.50 s0 %

HAT

OPENSIZZLECHICK

100 %0 %50 %

CHOKE

HIT

STRIKESTICK

100 %20 %

EQ

CUTAIR

150 Hz80 %

ERA

GRITROOMWIDTH

0 %0 %65 %

ECHO

ECHOTIME

0 %1/8D120 BPM

TRANSIENT

ATTACKSUSTAIN

0 %0 %

DRIVE

ENGINEDRIVECOLOUR

IDLE BURNHYPERDRIVERAZOR WINGSUPERNOVA

0 %100 %

COMP

COMPSPEED

0 %50 %

OUT

LEVELVELO

-12.9 dB50 %

KEYSGM KIT

WHAT IS IN HERE

03	QUICK START	Install it, hit it, pick a preset.
04	THE IDEA	Bronze, a circuit, and a wash.
05	METAL	The cymbal itself: eight knobs.
06	HAT	Two plates on a pedal.
07	HIT & EQ	Where the stick lands; what the mic hears.
08	ERA & ECHO	The machine, the room, the width, the dub.
09	TRANSIENT & DRIVE	Front and back; four engines.
10	COMP & OUT	Roll or pump; level, touch and keys.
11	THE DISPLAY & THE PAD	The hit, its brightness, a cymbal from above.
12	RECIPES	From one style to another.
13	PRESETS	Forty-four, in six banks.
15	SPECIFICATIONS	And what the bench measured.

IN SHORT

Hats Off synthesises hi-hats and cymbals: the real ones in a studio, the drum machines that made the records, and what techno, dub and dubstep did with them next.

It does not carry a simulation per machine. It models what every cymbal is — a plate that rings, closed or free — and the circuit that imitated it, with thirty-one controls, and lets you walk from one to the next.

THE FIRST TEN SECONDS

Open it and click the cymbal in the corner — the bell in the middle, then out towards the edge. Then **CLOSED**, **PEDAL** and **OPEN**. Press > beside the preset name until something makes you grin.

INSTALL, HIT, PLAY



The panel. The display and the cymbal pad across the top. Below, the controls in the order the sound meets them: Metal, Hat, Hit, EQ, then Era, Echo, Transient, Drive, Comp and Out.

1 • INSTALL

Copy **Hats Off.vst3** into C:\Program Files\Common Files\VST3\ and rescan. It files under Instrument / Drum. The standalone, **Hats Off.exe**, needs no installing.

2 • FIRST HIT

Click the cymbal: the middle is the bell, the rim is the edge. Keys: **space** as dialled, **C** closed, **P** pedal, **O** open, **B** bell, **E** edge.

3 • PICK A PRESET

< and > step through the banks; the name opens the list. Presets are level-matched, so you compare cymbals, not volumes.

4 • FROM MIDI

With **KEYS** on GM KIT (the default), 42 is the closed hat, 44 the pedal, 46 the open hat, 53 the bell, and 49, 52, 55 and 57 the edge — the General MIDI drum map, so a drum rack just works.

5 • KEEP IT

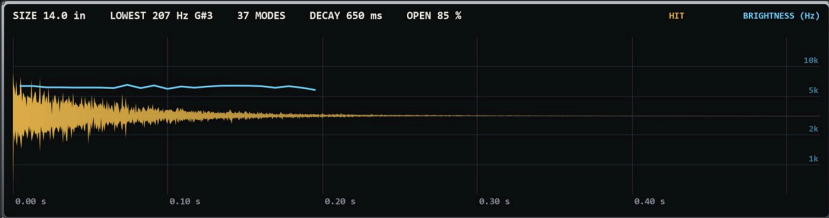
SAVE and **LOAD** use Documents\Brokild patches\Hats Off.

BRONZE, A CIRCUIT, A WASH

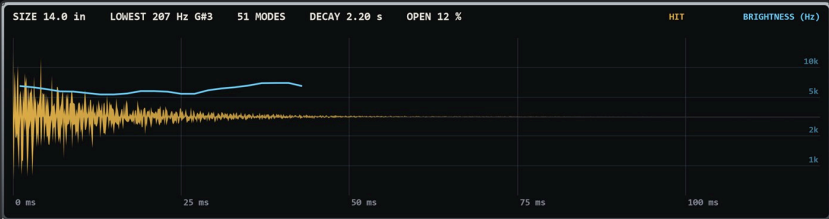
A cymbal is a plate of bronze with a great many modes that do not line up. A drum machine faked that with square waves. Hats Off builds both, and lets one turn into the other.

WHAT	808 / 606	A REAL CYMBAL	HATS OFF
The tone	six squares, not in tune	a plate’s modes	BRONZE walks between them
The top	a band-pass	too dense to hear as notes	DENSITY: modes, then a wash
Where it is hit	—	bell, bow, edge	STRIKE
The build	—	brighter after the hit	BLOOM
Open or closed	two sounds	one pedal	OPEN, CHICK, CHOKE
The stick	—	a tick on the ride	STICK

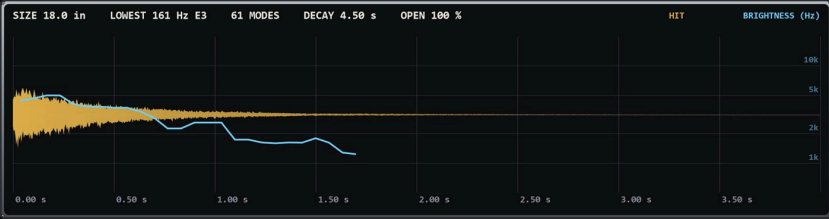
The plate’s low and middle modes are rung one by one, because that is where a cymbal’s pitch lives: a 14-inch and a 20-inch are different instruments. Above a few kHz a real cymbal’s modes crowd and couple until no ear hears them as notes, so there Hats Off lays a **wash** of noise bands that obey the same laws — the same decay, the same tilt with the strike, the same damping as the hat closes.



808 Open Hat. Six squares through a band-pass.



Studio Hi-Hat 14. Bronze, nearly closed.



Crash 18. The explosion, then the bloom.

METAL

PITCH ± 24 semitones

Moves every mode and every square together, so the cymbal keeps its character and changes its register.

SIZE 8-24 inches

The diameter. The lowest mode falls as the plate grows: a splash is high and quick, a 22-inch ride low and dark. The display names it.

BRONZE circuit $\rightarrow$ plate

At 0, the 808's six square waves through a band-pass. At 100, the plate. Between, both — the 909's trick.

DENSITY

How many modes, from ten to sixty-four, and how much wash above them. Low is a bell-like ping; high is the continuum of a crash.

BLOOM

A struck cymbal gets brighter for a moment after the stick, as the energy climbs into the high modes. BLOOM sets how much and how slowly: the swell of a crash.

TRASH

The plate rattles against its own lowest modes: a china, an upturned cymbal, a cracked one.

DECAY 30 ms-8 s

Time for the lowest mode to fall 60 dB with the hat fully open. Higher modes die sooner, as on a real plate.

NOISE

Plain white noise under the envelope: a CR-78, a 606, or grit for a machine hat.

HAT

Two plates on a pedal. The foot decides how long they ring.

OPEN

How far apart the plates are. Closed, the upper plate damps the lower and the high modes die first, so a closed hat is short and dark; fully open is a free cymbal ringing for DECAY.

SIZZLE

Half-open plates touch and buzz against each other. The rattle follows the plate's own motion, and it cannot happen when the hat is shut.

CHICK

The pedal note: the two plates clapped together by the foot, a short, woody chick at about 2.4 kHz.

CHOKE OFF / ON

With CHOKE on, a closed hit stops an open hat ringing — the foot coming down. Off, they overlap like two separate cymbals.

WHAT OPEN DOES, MEASURED

From fully open to fully closed, the time for a hit to fall 40 dB goes from **1339 ms** to **38 ms**. A closed hit after an open one leaves the open hat **113 dB** down 400 ms later with CHOKE, against 28 dB without.

THREE WAYS TO PLAY THE HAT

CLOSED clamps OPEN to a few percent; **PEDAL** is the chick on its own with a little plate; **OPEN** opens it to at least 80 %. All three from the same patch, so a preset is a whole pair of hats, not one sound.



909 Closed Hat. Bronze through six bits, choked short.

HIT & EQ

STRIKE bell → edge

Where the stick lands. On the **bell** the dome rings: a clear ping that lasts. On the **bow**, the ride. On the **edge** the high modes explode: the crash, brighter and shorter.

STICK

The tip of the stick on the metal: the tick that defines a ride pattern over the wash.

CUT 20 Hz–8 kHz

A high-pass, as every engineer puts on the overheads. At its bottom it is not there at all.

AIR

The top octave. Down for a dark, dusty cymbal; fully up is no filter.

SIX WAYS TO HIT IT

ARTICULATION	GM NOTE	PAD / KEY	WHAT CHANGES
As dialled	any other	cymbal · space	as the controls say
Closed	42	CLOSED · C	the hat shut
Pedal	44	PEDAL · P	the chick, the foot
Open	46	OPEN · O	at least 80 % open
Bell	53	middle · B	STRIKE on the bell
Edge	49 52 55 57	rim · E	STRIKE on the edge

Measured from bell to edge the brightness (spectral centroid) rises from **4.3 kHz** to **8.5 kHz**, and the bell rings on for **3.9 s** against 2.7 s for the edge.



Ride Bell. The dome: bright from the first moment, and long.

ERA & ECHO

ERA • THE MACHINE, THE ROOM, THE WIDTH

GRIT

Lower sample rate and bit depth. A little is the 12-bit SP-1200; more is the 909's six-bit cymbal ROM and the 8-bit Linn.

ROOM

A room around the cymbal — the overheads rather than the close mic. It sits before the drive and the compressor.

WIDTH

A cymbal is wide: its modes radiate to different sides. WIDTH spreads them across the stereo field and decorrelates the wash. At 0 the cymbal is mono, both sides the same sample for sample.

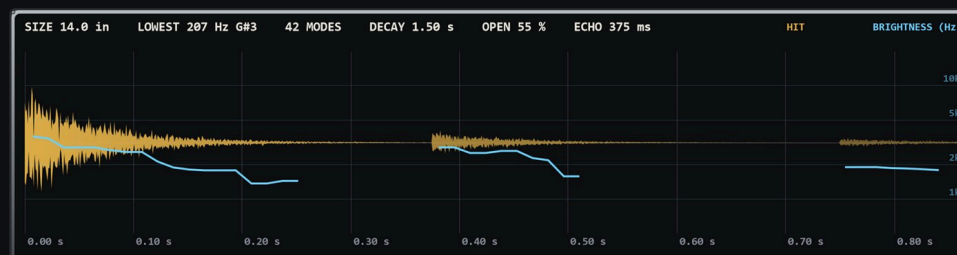
ECHO • THE DUB

ECHO

A tape echo, ping-pong: the first repeat on the left, the next on the right. One knob raises the send and the feedback together. Each pass comes back darker and thinner, and saturates, so it can be pushed to the top without running away.

TIME $1/16$... $1/2$

Locked to the host tempo, which is shown beneath it; 120 BPM when the standalone has no host.



Dub Hat Echo. A half-open hat and its repeats on a dotted eighth.

TRANSIENT & DRIVE

TRANSIENT

ATTACK bipolar

More or less of the first milliseconds: the stick.

SUSTAIN bipolar

More or less of the ring: the wash and the room.

A transient shaper usually has to guess from the audio. Hats Off is the synthesiser, so it reads its own envelopes — plate, stick and chick — and the gain it makes cannot ripple.

DRIVE • FOUR ENGINES FROM BATTLESTAR OVERDRIVE

ENGINE	WHAT IT DOES
IDLE BURN	A warm, asymmetric tube. Thickens without shouting.
HYPERDRIVE	Harmonic injection: a techno hat that cuts on small speakers.
RAZOR WING	Tanh into a hard clip. Hard techno, gabber.
SUPERNOVA	Fold, crush and chaos. Industrial sheet metal.

DRIVE

How hard the engine is pushed. Level-matched by measurement: more drive is more character, not more volume.

COLOUR

The drive's low-pass. Fully open is no filter at all.

A cymbal is nearly all top end, so the drive runs at eight times the sample rate: its harmonics land above hearing and are filtered away instead of folding back.

COMP & OUT

COMP

COMP

Threshold and ratio on one knob, with its own make-up, so turning it up makes the cymbal denser, not quieter.

SPEED

Slow lets the stick through before the compressor grabs the wash. Fast flattens it and brings the ring up to meet the hit — the pumping open hat. It looks 1.5 ms ahead, so even fast it catches the front.

ROLLS AND WASHES

Six cymbals ring at once, and they overlap the way real ones do. A seventh fades the oldest rather than cutting it, so a trap roll or sixteen crashes in a row stay smooth.

OUT

LEVEL

Output. A soft ceiling holds everything under 0 dBFS.

VELO

How much harder hits are louder and brighter. At 0, every velocity is the same hit, exactly.

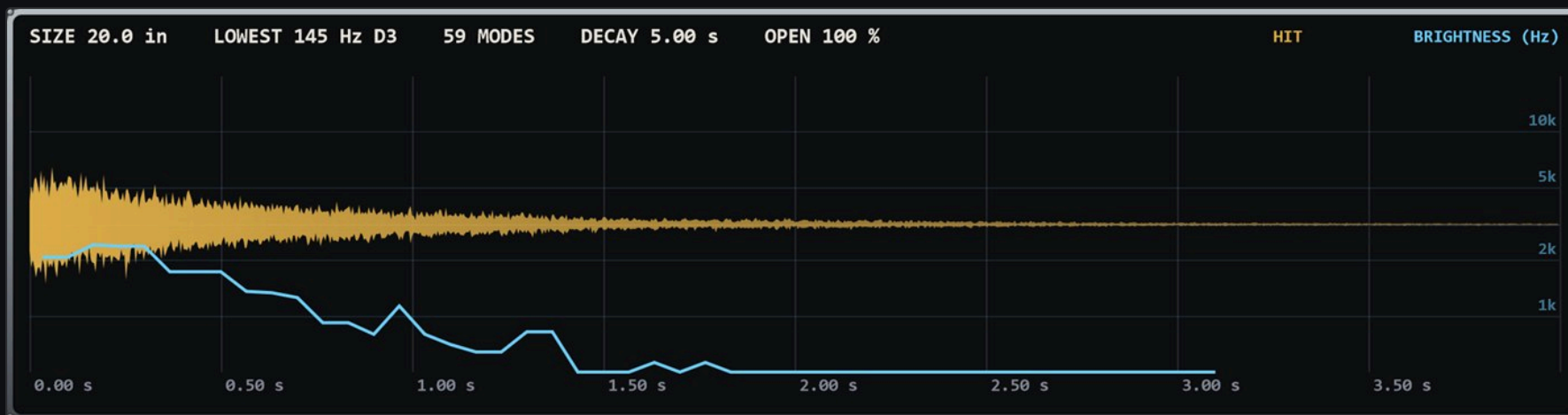
KEYS

FIXED: every note plays the cymbal as dialled. **GM KIT:** 42 closed, 44 pedal, 46 open, 53 bell, 49 / 52 / 55 / 57 edge. **CHROMATIC:** the note tunes the cymbal, with F#1 (42) at PITCH — a tuned bell line, or a gong played in the key.

NO MACHINE GUN

No two hits are the same: each mode's amplitude moves a little every time, as the stick never lands in the same place twice.

THE DISPLAY & THE PAD



Jazz Ride 20. The hit in brass, its brightness over time in steel; the readouts along the top.

THE HIT

Rendered by the same engine every time a control moves. OPEN is a hit that stops; DECAY one that does not.

ITS BRIGHTNESS

The steel curve is the spectral centroid over time. BLOOM shows as a curve that rises after the stick; a closed hat as one that falls at once.

PAD & METERS

A cymbal seen from above: click the bell, the bow or the edge. CLOSED, PEDAL and OPEN below it; OUT and GR meters beside.

RECIPES

From one style to another, a few knobs at a time.

808 → REAL HAT

1. Start on **808 Closed Hat**.
2. BRONZE up to 100 %.
3. DENSITY 70 %, SIZE 14 inches.
4. STICK and CHICK up; a little ROOM.

CLOSED → SIZZLING

1. A studio hat, OPEN 15 %.
2. SIZZLE up to 70 %.
3. Walk OPEN up to 35 %: the plates touch.
4. Automate OPEN: that is the foot.

RIDE → CRASH

1. Start on **Jazz Ride 20**.
2. SIZE down to 18, STRIKE to the edge.
3. BLOOM up to 70 %, STICK down.
4. DECAY to 4 s. It explodes now.

DUB HAT

1. A half-open hat, OPEN 30 %.
2. ECHO up, TIME 1/8D.
3. AIR down a touch, COLOUR down.
4. Ride ECHO from the host on the last beat.

HARD TECHNO

1. BRONZE 50 %: circuit and plate.
2. CUT to 3 kHz, NOISE 20 %.
3. RAZOR WING, DRIVE 60 %.
4. COMP 70 %, SPEED fast. Roll it.

THE CHINA

1. Any crash.
2. TRASH up to 70 %.
3. STRIKE to the edge, DENSITY up.
4. A little SUPERNOVA for a cracked one.

PRESETS

ACOUSTIC

Studio Hi-Hat 14

A pair of 14-inch hats, mic'd close: closed, pedal and open.

Loose Hat

The pedal eased off: the plates touch and sizzle.

Jazz Ride 20 • Sizzle Ride

A stick over a wash that builds — and one with rivets.

Ride Bell • Dark Ride 22

The ping; and a big, dark, dry ride.

Crash 16 • Crash 18

Thin and fast; the explosion, then the bloom.

Splash 10 • China 18

Short and bright; trashy, upturned, rude.

Hats In A Room

The hi-hat heard from the overheads.

Init

A 14-inch hat, nearly closed: a place to start.

VINTAGE

808 Closed • Open • Cymbal

Six squares and a band-pass; the closed one chokes the open.

909 Closed • Open • Ride • Crash

A real cymbal through six bits — the house open hat.

606 Hats

Squares and noise, thin and ticking.

CR-78 Hat

A burst of noise, politely filtered.

LinnDrum • 707 / 727

A real hat, 8-bit; the mid-eighties Roland.

SP-1200 Hat

12-bit, filtered, pushed.

Kraftwerk Metal

A ring-modulated circuit cymbal with an echo.

TECHNO

Techno Closed

Circuit, noise and bronze through HYPERDRIVE.

Rolling Open Hat

The off-beat open hat, compressed to roll.

Minimal Tick

A tiny, wide, dry tick.

Detroit Ride

A gritty machine ride.

Hard Techno Hat

RAZOR WING and a compressor.

Warehouse Ride

A ride in a big concrete room.

PRESETS

DUB

Dub Hat Echo

A half-open hat into a dotted-eighth tape echo.

Roots Hi-Hat

A loose real hat with a sixteenth-note echo.

Dub Techno Hat

Dark, open; the echo does the rest.

Skank Hat

A closed hat with an eighth-note echo.

Space Cymbal

A crash thrown into the echo and the room.

DUBSTEP

Dubstep Hat

Tight, bright, compressed, all attack.

Halftime Crash

A big crash in a room for the drop.

Trap Hat

A short hat for rolls; velocity does the rest.

Grime Hat

A crushed circuit hat.

UKG Shuffle Hat

A half-open hat for the shuffle.

MODERN

House Open Hat

The open hat on the off-beat.

Lo-Fi Hat

A hat left in a drawer since 1994.

Metal Plate

A 24-inch sheet of trash through SUPERNOVA.

Psytrance Hat

A bright, wide, half-open hat.

LEVEL-MATCHED

Every preset is levelled by measurement: –17 dB RMS over the first 80 ms, the part of a cymbal the ear judges its loudness by, and no clean preset above –1 dBFS — so flicking through compares cymbals, not volumes. The machine names are in the spirit of the originals; nothing is sampled from them.

SPECIFICATIONS

Format	Windows VST3, 64-bit, and a standalone application
Type	Instrument / Drum. MIDI in, stereo out
Sound	Fully synthesised. No samples
Controls	31: Metal, Hat, Hit, EQ, Era, Echo, Transient, Drive, Comp, Out
Articulations	As dialled, closed, pedal, open, bell, edge
Keys	FIXED, GM KIT (42 / 44 / 46 / 53 / 49), CHROMATIC
Polyphony	Six ringing cymbals; the oldest fades
The plate	Up to 64 modes and 8 bell partials, plus an 8-band wash
Drive engines	IDLE BURN, HYPERDRIVE, RAZOR WING, SUPERNOVA
Presets	44 factory in six banks, plus Init
User patches	Documents\Brokild patches\Hats Off
Licence	AGPLv3. Source at github.com/peterboggild/BrokildApps
Build	260926.4

MEASURED

Lowest mode against SIZE	within 0.001 cents
DECAY against its -60 dB time	within 0.3 %
Oversampling	4x; drive at 8x
Aliasing, RAZOR WING at 100 %	-84 dB
Open hat after a choke, 400 ms on	-113 dB
Echo, 1/8D at 120 BPM	375 ms, left then right
Drive level change	within 1.4 dB
Latency at 48 kHz	103 samples, 2.1 ms
CPU, a ride with six ringing	about 11 % of a core

Latency is reported to the host and compensated. Every number here comes from the bench in the source, which renders real audio and measures it: 53 engine checks, 76 in a plugin host, 12 on the panel.

CREDITS

HATS OFF TO THE DRUMMERS

This VST comes from the curved mind of Professor Brokild, and was programmed with VScode, ChatGPT and Claude. Use of this is on your own peril. It may generate unique, beautiful, scary or useless sounds for you, and it may crash your DAW project. Have fun, play unsafely, unwisely, and unboringly.

It finishes a trilogy with **Kickstart**, the kick drum synthesiser, and **Snare Tactics**, the snare. The demos on the website play all three together.

Cheers, Peter Bøggild, September 2026.

Drum machine names are trademarks of their owners, used only to describe the sound a preset is in the spirit of. Hats Off contains no samples of them.

BROKILD

Free, and free to pass on.
peterboggild.github.io/BrokildApps

BUILD 260926.4 • WINDOWS VST3

